FAIRCHILD
SEMICONDUCTOR™

August 1986
Revised March 2000

DM74LS00
Quad 2-Input NAND Gate

General Description

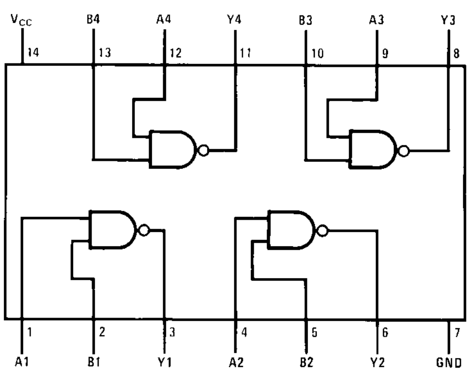
This device contains four independent gates each of which performs the logic NAND function.

Ordering Code:

Order Number	Package Number	Package Description
DM74LS00M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS00SJ	M14D	14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS00N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Connection Diagram



Function Table

$Y = \overline{AB}$

Inputs		Output
A	**B**	**Y**
L	L	H
L	H	H
H	L	H
H	H	L

H = HIGH Logic Level
L = LOW Logic Level

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May 1986
Revised March 2000

DM74LS02
Quad 2-Input NOR Gate

General Description

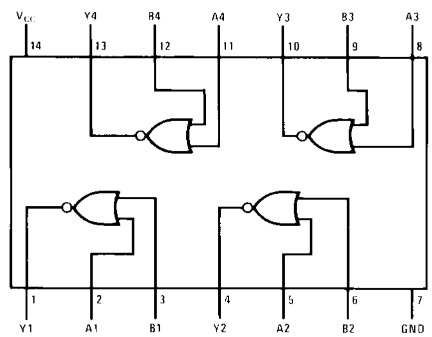
This device contains four independent gates each of which performs the logic NOR function.

Ordering Code:

Order Number	Package Number	Package Description
DM74LS02M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS02SJ	M14D	14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS02N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Connection Diagram



Function Table

$$Y = \overline{A + B}$$

Inputs		Output
A	B	Y
L	L	H
L	H	L
H	L	L
H	H	L

H = HIGH Logic Level
L = LOW Logic Level

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August 1986
Revised March 2000

DM74LS04
Hex Inverting Gates

General Description

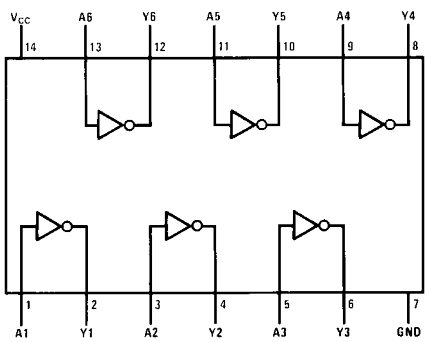
This device contains six independent gates each of which performs the logic INVERT function.

Ordering Code:

Order Number	Package Number	Package Description
DM74LS04M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS04SJ	M14D	14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS04N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code.

Connection Diagram



Function Table

$Y = \overline{A}$

Input	Output
A	**Y**
L	H
H	L

H = HIGH Logic Level
L = LOW Logic Level

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August 1986
Revised March 2000

DM74LS08
Quad 2-Input AND Gates

General Description

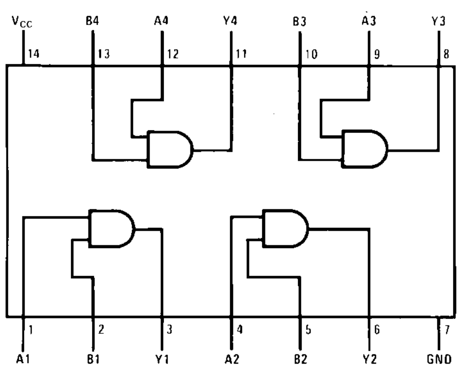
This device contains four independent gates each of which performs the logic AND function.

Ordering Code:

Order Number	Package Number	Package Description
DM74LS08M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS08SJ	M14D	14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS08N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code.

Connection Diagram



Function Table

Y = AB

Inputs		Output
A	**B**	**Y**
L	L	L
L	H	L
H	L	L
H	H	H

H = HIGH Logic Level
L = LOW Logic Level

 DS006347

FAIRCHILD
SEMICONDUCTOR™

June 1986
Revised March 2000

DM74LS32
Quad 2-Input OR Gate

General Description

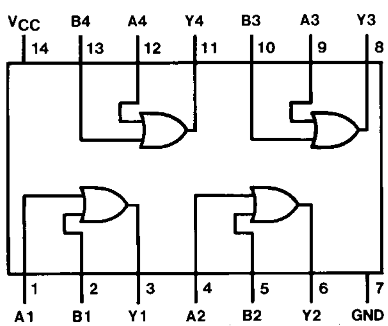
This device contains four independent gates each of which performs the logic OR function.

Ordering Code:

Order Number	Package Number	Package Description
DM74LS32M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS32SJ	M14D	14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS32N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Connection Diagram



Function Table

Y = A + B

Inputs		Output
A	B	Y
L	L	L
L	H	H
H	L	H
H	H	H

H = HIGH Logic Level
L = LOW Logic Level

 DS006361

FAIRCHILD
SEMICONDUCTOR TM

October 1988
Revised March 2000

DM74LS47
BCD to 7-Segment Decoder/Driver with Open-Collector Outputs

General Description

The DM74LS47 accepts four lines of BCD (8421) input data, generates their complements internally and decodes the data with seven AND/OR gates having open-collector outputs to drive indicator segments directly. Each segment output is guaranteed to sink 24 mA in the ON (LOW) state and withstand 15V in the OFF (HIGH) state with a maximum leakage current of 250 μA. Auxiliary inputs provided blanking, lamp test and cascadable zero-suppression functions.

Features

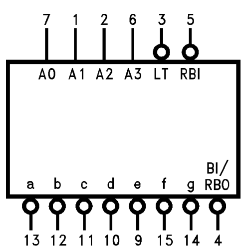
- Open-collector outputs
- Drive indicator segments directly
- Cascadable zero-suppression capability
- Lamp test input

Ordering Code:

Order Number	Package Number	Package Description
DM74LS47M	M16A	16-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150 Narrow
DM74LS47N	N16E	16-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

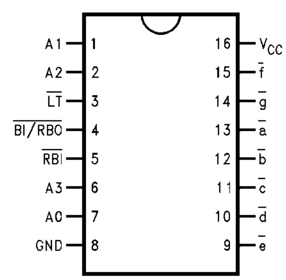
Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Logic Symbol



V_{CC} = Pin 16
GND = Pin 8

Connection Diagram



Pin Descriptions

Pin Names	Description
A0–A3	BCD Inputs
$\overline{RBI}$	Ripple Blanking Input (Active LOW)
$\overline{LT}$	Lamp Test Input (Active LOW)
$\overline{BI}/\overline{RBO}$	Blanking Input (Active LOW) or Ripple Blanking Output (Active LOW)
$\overline{a}$ –$\overline{g}$	Segment Outputs (Active LOW) (Note 1)

Note 1: OC—Open Collector

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August 1986
Revised March 2000

DM74LS86
Quad 2-Input Exclusive-OR Gate

General Description

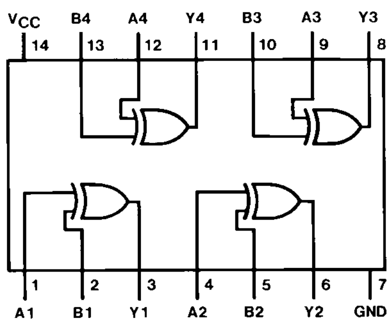
This device contains four independent gates each of which performs the logic exclusive-OR function.

Ordering Code:

Order Number	Package Number	Package Description
DM74LS86M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS86SJ	M14D	14-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS86N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Connection Diagram



Function Table

$$Y = A \oplus B = \overline{A}\,B + A\overline{B}$$

Inputs		Output
A	**B**	**Y**
L	L	L
L	H	H
H	L	H
H	H	L

H = HIGH Logic Level
L = LOW Logic Level

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August 1986
Revised March 2000

DM74LS90
Decade and Binary Counters

General Description

Each of these monolithic counters contains four master-slave flip-flops and additional gating to provide a divide-by-two counter and a three-stage binary counter for which the count cycle length is divide-by-five for the DM74LS90.

All of these counters have a gated zero reset and the DM74LS90 also has gated set-to-nine inputs for use in BCD nine's complement applications.

To use their maximum count length (decade or four bit binary), the B input is connected to the Q_A output. The input count pulses are applied to input A and the outputs are as described in the appropriate truth table. A symmetrical divide-by-ten count can be obtained from the DM74LS90 counters by connecting the Q_D output to the A input and applying the input count to the B input which gives a divide-by-ten square wave at output Q_A.

Features

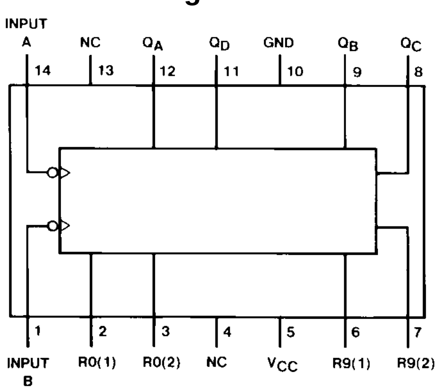
- Typical power dissipation 45 mW
- Count frequency 42 MHz

Ordering Code:

Order Number	Package Number	Package Description
DM74LS90M	M14A	14-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-120, 0.150 Narrow
DM74LS90N	N14A	14-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code.

Connection Diagram



Reset/Count Truth Table

Reset Inputs				Output			
R0(1)	R0(2)	R9(1)	R9(2)	Q_D	Q_C	Q_B	Q_A
H	H	L	X	L	L	L	L
H	H	X	L	L	L	L	L
X	X	H	H	H	L	L	H
X	L	X	L	COUNT			
L	X	L	X	COUNT			
L	X	X	L	COUNT			
X	L	L	X	COUNT			

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August 1986
Revised March 2000

DM74LS138 • DM74LS139 Decoder/Demultiplexer

General Description

These Schottky-clamped circuits are designed to be used in high-performance memory-decoding or data-routing applications, requiring very short propagation delay times. In high-performance memory systems these decoders can be used to minimize the effects of system decoding. When used with high-speed memories, the delay times of these decoders are usually less than the typical access time of the memory. This means that the effective system delay introduced by the decoder is negligible.

The DM74LS138 decodes one-of-eight lines, based upon the conditions at the three binary select inputs and the three enable inputs. Two active-low and one active-high enable inputs reduce the need for external gates or inverters when expanding. A 24-line decoder can be implemented with no external inverters, and a 32-line decoder requires only one inverter. An enable input can be used as a data input for demultiplexing applications.

The DM74LS139 comprises two separate two-line-to-four-line decoders in a single package. The active-low enable input can be used as a data line in demultiplexing applications.

All of these decoders/demultiplexers feature fully buffered inputs, presenting only one normalized load to its driving circuit. All inputs are clamped with high-performance Schottky diodes to suppress line-ringing and simplify system design.

Features

- Designed specifically for high speed:
  - Memory decoders
  - Data transmission systems
- DM74LS138 3-to-8-line decoders incorporates 3 enable inputs to simplify cascading and/or data reception
- DM74LS139 contains two fully independent 2-to-4-line decoders/demultiplexers
- Schottky clamped for high performance
- Typical propagation delay (3 levels of logic)
  - DM74LS138 21 ns
  - DM74LS139 21 ns
- Typical power dissipation
  - DM74LS138 32 mW
  - DM74LS139 34 mW

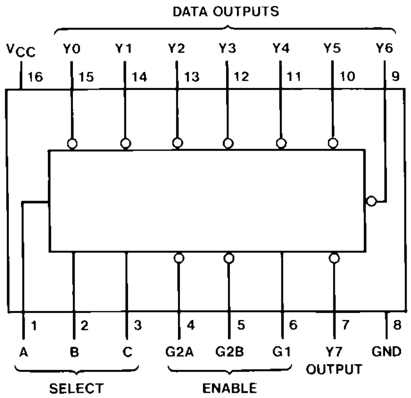
Ordering Code:

Order Number	Package Number	Package Description
DM74LS138M	M16A	16-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150 Narrow
DM74LS138SJ	M16D	16-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS138N	N16E	16-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide
DM74LS139M	M16A	16-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-012, 0.150 Narrow
DM74LS139SJ	M16D	16-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS139N	N16E	16-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

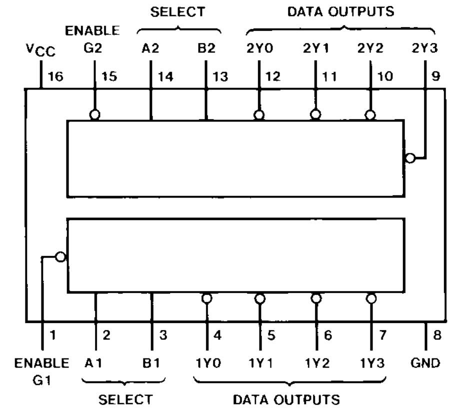
Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Connection Diagrams

DM74LS138



DM74LS139



Function Tables

DM74LS138

Inputs					Outputs							
Enable		Select										
G1	G2 (Note 1)	C	B	A	YO	Y1	Y2	Y3	Y4	Y5	Y6	Y7
X	H	X	X	X	H	H	H	H	H	H	H	H
L	X	X	X	X	H	H	H	H	H	H	H	H
H	L	L	L	L	L	H	H	H	H	H	H	H
H	L	L	L	H	H	L	H	H	H	H	H	H
H	L	L	H	L	H	H	L	H	H	H	H	H
H	L	L	H	H	H	H	H	L	H	H	H	H
H	L	H	L	L	H	H	H	H	L	H	H	H
H	L	H	L	H	H	H	H	H	H	L	H	H
H	L	H	H	L	H	H	H	H	H	H	L	H
H	L	H	H	H	H	H	H	H	H	H	H	L

DM74LS139

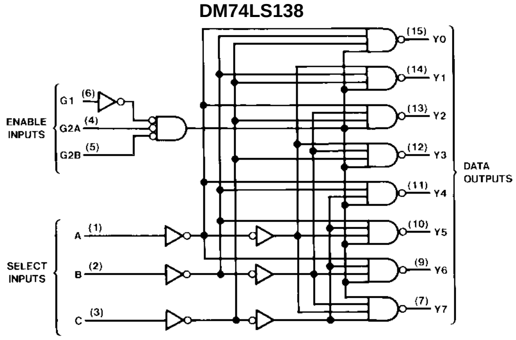
Inputs			Outputs			
Enable	Select					
G	B	A	Y0	Y1	Y2	Y3
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	L	H	H	L	H	H
L	H	L	H	H	L	H
L	H	H	H	H	H	L

H = HIGH Level
L = LOW Level
X = Don't Care

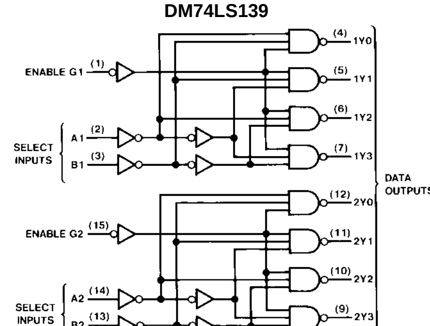
Note 1: G2 = G2A + G2B

Logic Diagrams

DM74LS138



DM74LS139



MOTOROLA

BCD DECADE COUNTERS/ 4-BIT BINARY COUNTERS

SN54/74LS160A
SN54/74LS161A
SN54/74LS162A
SN54/74LS163A

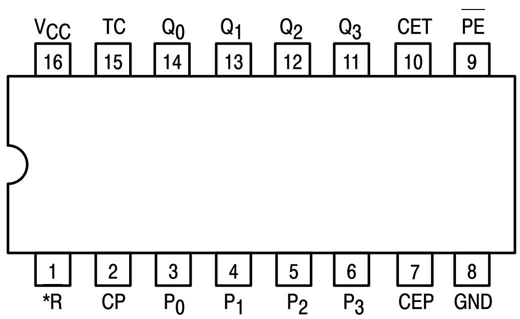
The LS160A/161A/162A/163A are high-speed 4-bit synchronous counters. They are edge-triggered, synchronously presettable, and cascadable MSI building blocks for counting, memory addressing, frequency division and other applications. The LS160A and LS162A count modulo 10 (BCD). The LS161A and LS163A count modulo 16 (binary.)

The LS160A and LS161A have an asynchronous Master Reset (Clear) input that overrides, and is independent of, the clock and all other control inputs. The LS162A and LS163A have a Synchronous Reset (Clear) input that overrides all other control inputs, but is active only during the rising clock edge.

	BCD (Modulo 10)	Binary (Modulo 16)
Asynchronous Reset	LS160A	LS161A
Synchronous Reset	LS162A	LS163A

- Synchronous Counting and Loading
- Two Count Enable Inputs for High Speed Synchronous Expansion
- Terminal Count Fully Decoded
- Edge-Triggered Operation
- Typical Count Rate of 35 MHz
- ESD > 3500 Volts

CONNECTION DIAGRAM DIP (TOP VIEW)



NOTE:
The Flatpak version has the same pinouts (Connection Diagram) as the Dual In-Line Package.

*MR for LS160A and LS161A
*SR for LS162A and LS163A

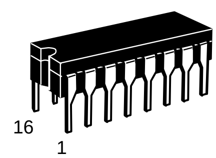
PIN NAMES

		LOADING (Note a) HIGH	LOW
$\overline{PE}$	Parallel Enable (Active LOW) Input	1.0 U.L.	0.5 U.L.
P_0-P_3	Parallel Inputs	0.5 U.L.	0.25 U.L.
CEP	Count Enable Parallel Input	0.5 U.L.	0.25 U.L.
CET	Count Enable Trickle Input	1.0 U.L.	0.5 U.L.
CP	Clock (Active HIGH Going Edge) Input	0.5 U.L.	0.25 U.L.
$\overline{MR}$	Master Reset (Active LOW) Input	0.5 U.L.	0.25 U.L.
$\overline{SR}$	Synchronous Reset (Active LOW) Input	1.0 U.L.	0.5 U.L.
Q_0-Q_3	Parallel Outputs (Note b)	10 U.L.	5 (2.5) U.L.
TC	Terminal Count Output (Note b)	10 U.L.	5 (2.5) U.L.

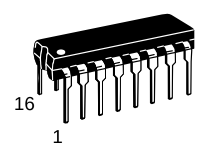
NOTES:
a) 1 TTL Unit Load (U.L.) = 40 µA HIGH/1.6 mA LOW.
b) The Output LOW drive factor is 2.5 U.L. for Military (54) and 5 U.L. for Commercial (74) Temperature Ranges.

BCD DECADE COUNTERS/ 4-BIT BINARY COUNTERS

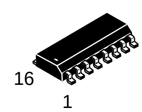
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

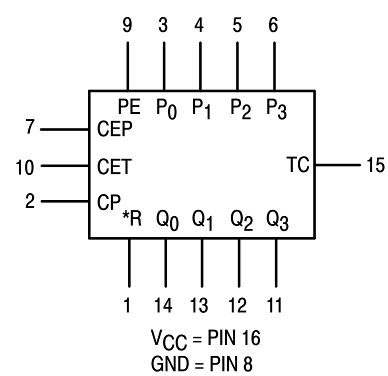


D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

SN54LSXXXJ	Ceramic
SN74LSXXXN	Plastic
SN74LSXXXD	SOIC

LOGIC SYMBOL



*MR for LS160A and LS161A
*SR for LS162A and LS163A

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August 1986
Revised March 2000

DM74LS245
3-STATE Octal Bus Transceiver

General Description

These octal bus transceivers are designed for asynchronous two-way communication between data buses. The control function implementation minimizes external timing requirements.

The device allows data transmission from the A Bus to the B Bus or from the B Bus to the A Bus depending upon the logic level at the direction control (DIR) input. The enable input ($\overline{G}$) can be used to disable the device so that the buses are effectively isolated.

Features

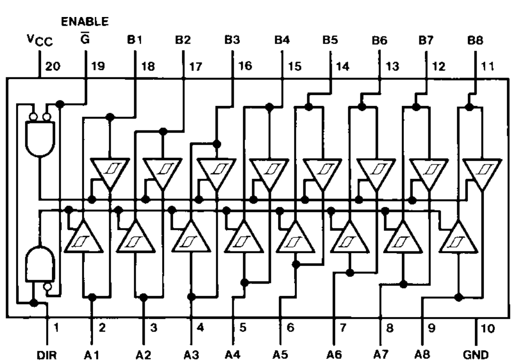
- Bi-Directional bus transceiver in a high-density 20-pin package
- 3-STATE outputs drive bus lines directly
- PNP inputs reduce DC loading on bus lines
- Hysteresis at bus inputs improve noise margins
- Typical propagation delay times, port-to-port 8 ns
- Typical enable/disable times 17 ns
- I_{OL} (sink current)
  24 mA
- I_{OH} (source current)
  –15 mA

Ordering Code:

Order Number	Package Number	Package Description
DM74LS245WM	M20B	20-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-013, 0.300 Wide
DM74LS245SJ	M20D	20-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS245N	N20A	20-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

Devices also available in Tape and Reel. Specify by appending the suffix letter "X" to the ordering code.

Connection Diagram



Function Table

Enable $\overline{G}$	Direction Control DIR	Operation
L	L	B Data to A Bus
L	H	A Data to B Bus
H	X	Isolation

H = HIGH Level
L = LOW Level
X = Irrelevant

FAIRCHILD
SEMICONDUCTOR™

October 1988
Revised March 2000

DM74LS273
8-Bit Register with Clear

General Description

The DM74LS273 is a high speed 8-bit register, consisting of eight D-type flip-flops with a common Clock and an asynchronous active LOW Master Reset. This device is supplied in a 20-pin package featuring 0.3 inch row spacing.

Features

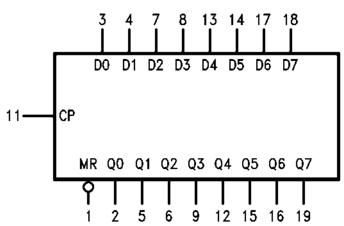
- Edge-triggered
- 8-bit high speed register
- Parallel in and out
- Common clock and master reset

Ordering Code:

Order Number	Package Number	Package Description
DM74LS273WM	M20B	20-Lead Small Outline Integrated Circuit (SOIC), JEDEC MS-013, 0.300 Wide
DM74LS273SJ	M20D	20-Lead Small Outline Package (SOP), EIAJ TYPE II, 5.3mm Wide
DM74LS273N	N20A	20-Lead Plastic Dual-In-Line Package (PDIP), JEDEC MS-001, 0.300 Wide

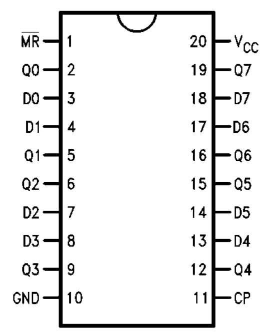
Devices also available in Tape and Reel. Specify by appending the suffix letter “X” to the ordering code.

Logic Symbol



V_{CC} = Pin 20
GND = Pin 10

Connection Diagram



Pin Descriptions

Pin Names	Description
CP	Clock Pulse Input (Active Rising Edge)
D0–D7	Data Inputs
$\overline{MR}$	Asynchronous Master Reset Input (Active LOW)
Q0–Q7	Flip-Flop Outputs

Truth Table

Inputs			Outputs
MR	CP	D_n	Q_n
L	X	X	L
H	↑	H	H
H	↑	L	L

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

CAT28C16A

16 kb CMOS Parallel EEPROM

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http://onsemi.com

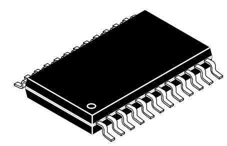
Description

The CAT28C16A is a fast, low power, 5V−only CMOS Parallel EEPROM organized as 2K x 8−bits. It requires a simple interface for in−system programming. On−chip address and data latches, self−timed write cycle with auto−clear and V_{CC} power up/down write protection eliminate additional timing and protection hardware. $\overline{\text{DATA}}$ Polling signals the start and end of the self−timed write cycle. Additionally, the CAT28C16A features hardware write protection.

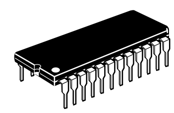
The CAT28C16A is manufactured using ON Semiconductor's advanced CMOS floating gate technology. It is designed to endure 100,000 program/erase cycles and has a data retention of 100 years. The device is available in JEDEC approved 24−pin DIP and SOIC or 32−pin PLCC packages.

Features

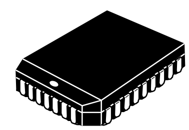
- Fast Read Access Times: 90 ns, 120 ns, 200 ns
- Low Power CMOS Dissipation:
  - Active: 25 mA Max.
  - Standby: 100 µA Max.
- Simple Write Operation:
  - On−chip Address and Data Latches
  - Self−timed Write Cycle with Auto−clear
- Fast Write Cycle Time: 10 ms Max
- End of Write Detection: $\overline{\text{DATA}}$ Polling
- Hardware Write Protection
- CMOS and TTL Compatible I/O
- 100,000 Program/Erase Cycles
- 100 Year Data Retention
- Commercial, Industrial and Automotive Temperature Ranges



SOIC−24
J, K, W, X SUFFIX
CASE 751BK



PDIP−24
L SUFFIX
CASE 646AD



PLCC−32
N, G SUFFIX
CASE 776AK

PIN FUNCTION

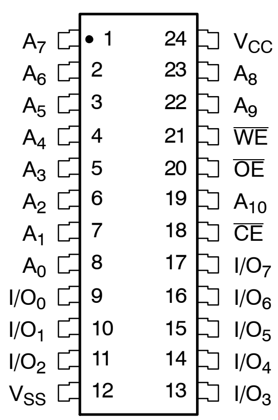
Pin Name	Function
A_0–A_{10}	Address Inputs
I/O_0–I/O_7	Data Inputs/Outputs
$\overline{\text{CE}}$	Chip Enable
$\overline{\text{OE}}$	Output Enable
$\overline{\text{WE}}$	Write Enable
V_{CC}	5 V Supply
V_{SS}	Ground
NC	No Connect

ORDERING INFORMATION

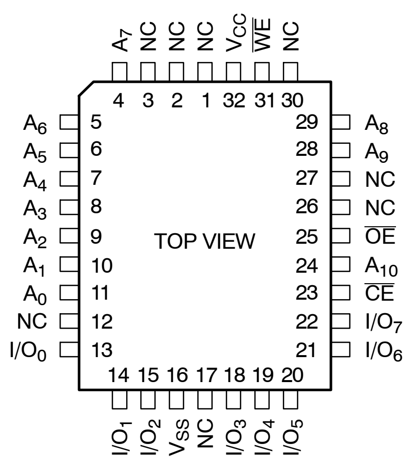
See detailed ordering and shipping information in the package dimensions section on page 10 of this data sheet.

PIN CONFIGURATION

DIP Package (L)
SOIC Package (J, K, W, X)



PLCC Package (N, G)



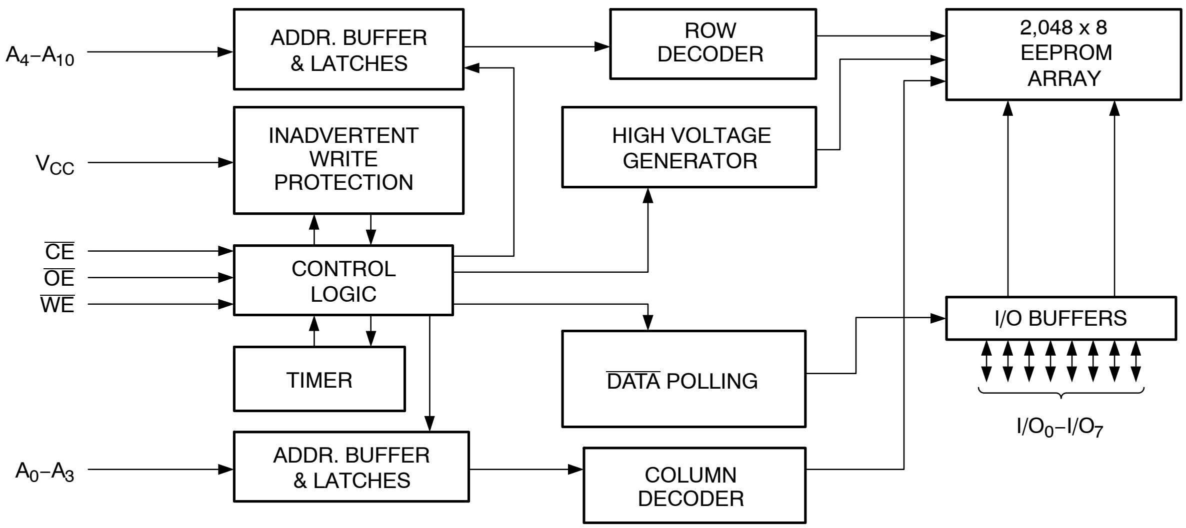


Figure 1. Block Diagram

Table 1. MODE SELECTION

Mode	$\overline{CE}$	$\overline{WE}$	$\overline{OE}$	I/O	Power
Read	L	H	L	D_{OUT}	ACTIVE
Byte Write ($\overline{WE}$ Controlled)	L		H	D_{IN}	ACTIVE
Byte Write ($\overline{CE}$ Controlled)		L	H	D_{IN}	ACTIVE
Standby, and Write Inhibit	H	X	X	High−Z	STANDBY
Read and Write Inhibit	X	H	H	High−Z	ACTIVE

Table 2. CAPACITANCE (T_A = 25°C, f = 1.0 MHz, V_{CC} = 5 V)

Symbol	Test	Max	Conditions	Units
$C_{I/O}$ (Note 1)	Input/Output Capacitance	10	$V_{I/O}$ = 0 V	pF
C_{IN} (Note 1)	Input Capacitance	6	V_{IN} = 0 V	pF

1. This parameter is tested initially and after a design or process change that affects the parameter.

Table 3. ABSOLUTE MAXIMUM RATINGS

Parameters	Ratings	Units
Temperature Under Bias	–55 to +125	°C
Storage Temperature	–65 to +150	°C
Voltage on Any Pin with Respect to Ground (Note 2)	–2.0 V to +V_{CC} + 2.0 V	V
V_{CC} with Respect to Ground	–2.0 to +7.0	V
Package Power Dissipation Capability (T_A = 25°C)	1.0	W
Lead Soldering Temperature (10 secs)	300	°C
Output Short Circuit Current (Note 3)	100	mA

Stresses exceeding Maximum Ratings may damage the device. Maximum Ratings are stress ratings only. Functional operation above the Recommended Operating Conditions is not implied. Extended exposure to stresses above the Recommended Operating Conditions may affect device reliability.

2. The minimum DC input voltage is −0.5 V. During transitions, inputs may undershoot to −2.0 V for periods of less than 20 ns. Maximum DC voltage on output pins is V_{CC} + 0.5 V, which may overshoot to V_{CC} + 2.0 V for periods of less than 20 ns.
3. Output shorted for no more than one second. No more than one output shorted at a time.

Table 4. RELIABILITY CHARACTERISTICS (Note 4)

Symbol	Parameter	Min	Max	Units
N_{END} (Note 5)	Endurance	100,000		Cycles/Byte
T_{DR} (Notes 5)	Data Retention	100		Years
V_{ZAP}	ESD Susceptibility	2,000		V
I_{LTH} (Note 6)	Latch−Up	100		mA

4. This parameter is tested initially and after a design or process change that affects the parameter.
5. For the CAT28C16A−20, the minimum endurance is 10,000 cycles and the minimum data retention is 10 years.
6. Latch−up protection is provided for stresses up to 100 mA on address and data pins from −1 V to V_{CC} + 1 V.

Table 5. D.C. OPERATING CHARACTERISTICS (V_{CC} = 5 V ±10%, unless otherwise specified.)

Symbol	Parameter	Test Conditions	Limits			Units
			Min	Typ	Max	
I_{CC}	V_{CC} Current (Operating, TTL)	$\overline{CE} = \overline{OE} = V_{IL}$, f = 1/$t_{RC}$ min, All I/O's Open			35	mA
I_{CCC} (Note 7)	V_{CC} Current (Operating, CMOS)	$\overline{CE} = \overline{OE} = V_{ILC}$, f = 1/$t_{RC}$ min, All I/O's Open			25	mA
I_{SB}	V_{CC} Current (Standby, TTL)	$\overline{CE} = V_{IH}$, All I/O's Open			1	mA
I_{SBC} (Note 8)	V_{CC} Current (Standby, CMOS)	$\overline{CE} = V_{IHC}$, All I/O's Open			100	μA
I_{LI}	Input Leakage Current	V_{IN} = GND to V_{CC}	−10		10	μA
I_{LO}	Output Leakage Current	V_{OUT} = GND to V_{CC}, $\overline{CE} = V_{IH}$	−10		10	μA
V_{IH} (Note 8)	High Level Input Voltage		2		V_{CC} + 0.3	V
V_{IL} (Note 7)	Low Level Input Voltage		−0.3		0.8	V
V_{OH}	High Level Output Voltage	I_{OH} = −400 μA	2.4			V
V_{OL}	Low Level Output Voltage	I_{OL} = 2.1 mA			0.4	V
V_{WI}	Write Inhibit Voltage		3.0			V

7. V_{ILC} = −0.3 V to +0.3 V
8. V_{IHC} = V_{CC} − 0.3 V to V_{CC} + 0.3 V

Table 6. A.C. CHARACTERISTICS, READ CYCLE (V_{CC} = 5 V ±10%, unless otherwise specified.)

Symbol	Parameter	28C16A−90		28C16A−12		28C16A−20		Units
		Min	Max	Min	Max	Min	Max	
t_{RC}	Read Cycle Time	90		120		200		ns
t_{CE}	$\overline{CE}$ Access Time		90		120		200	ns
t_{AA}	Address Access Time		90		120		200	ns
t_{OE}	$\overline{OE}$ Access Time		50		60		80	ns
t_{LZ} (Note 9)	$\overline{CE}$ Low to Active Output	0		0		0		ns
t_{OLZ} (Note 9)	$\overline{OE}$ Low to Active Output	0		0		0		ns
t_{HZ} (Notes 9, 10)	$\overline{CE}$ High to High−Z Output		50		50		55	ns
t_{OHZ} (Notes 9, 10)	$\overline{OE}$ High to High−Z Output		50		50		55	ns
t_{OH} (Note 9)	Output Hold from Address Change	0		0		0		ns

9. This parameter is tested initially and after a design or process change that affects the parameter.
10. Output floating (High−Z) is defined as the state when the external data line is no longer driven by the output buffer.

LMC555 CMOS Timer

Check for Samples: LMC555

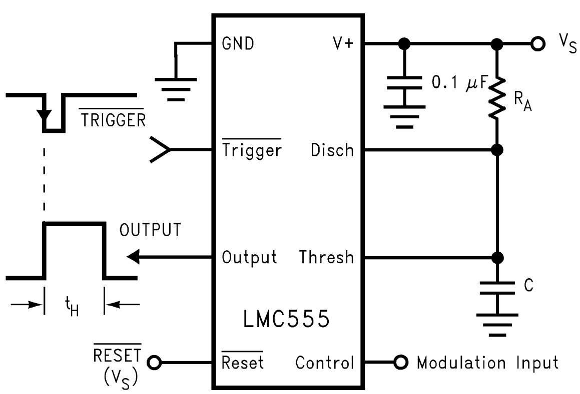
FEATURES

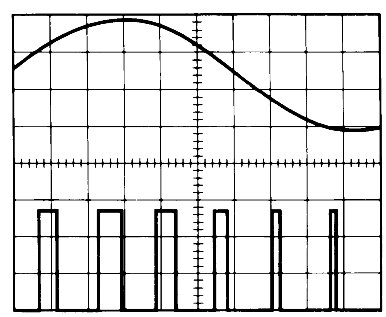
- **Less than 1 mW Typical Power Dissipation at 5V Supply**
- **3 MHz Astable Frequency Capability**
- **1.5V Supply Operating Voltage Ensured**
- **Output Fully Compatible with TTL and CMOS Logic at 5V Supply**
- **Tested to −10 mA, +50 mA Output Current Levels**
- **Reduced Supply Current Spikes During Output Transitions**
- **Extremely Low Reset, Trigger, and Threshold Currents**
- **Excellent Temperature Stability**
- **Pin-for-Pin Compatible with 555 Series of Timers**
- **Available in 8-pin VSSOP Package and 8-Bump DSBGA package**

DESCRIPTION

The LMC555 is a CMOS version of the industry standard 555 series general purpose timers. In addition to the standard package (SOIC, VSSSOP, and PDIP) the LMC555 is also available in a chip sized package (8 Bump DSBGA) using TI's DSBGA package technology. The LMC555 offers the same capability of generating accurate time delays and frequencies as the LM555 but with much lower power dissipation and supply current spikes. When operated as a one-shot, the time delay is precisely controlled by a single external resistor and capacitor. In the stable mode the oscillation frequency and duty cycle are accurately set by two external resistors and one capacitor. The use of Texas Instruments' LMCMOS process extends both the frequency range and low supply capability.

Pulse Width Modulator





Connection Diagram

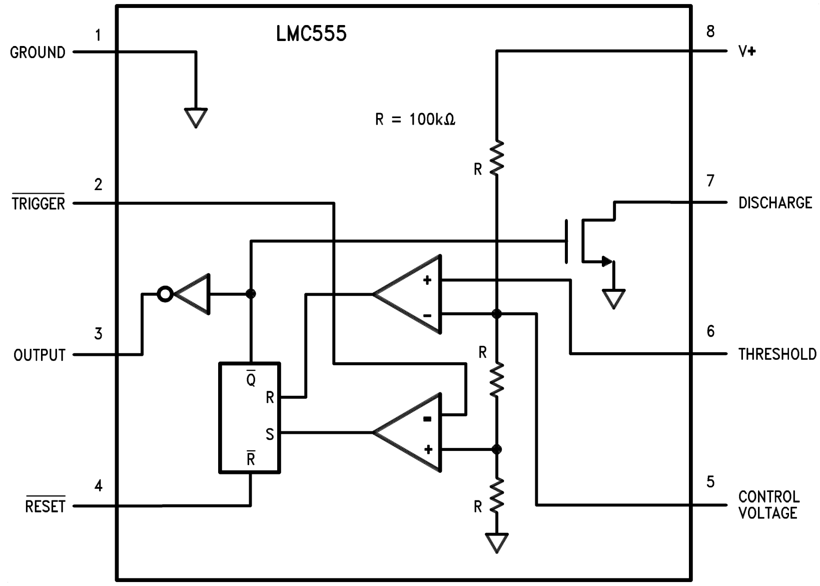


Figure 1. 8-Pin SOIC, VSSOP, PDIP Top View

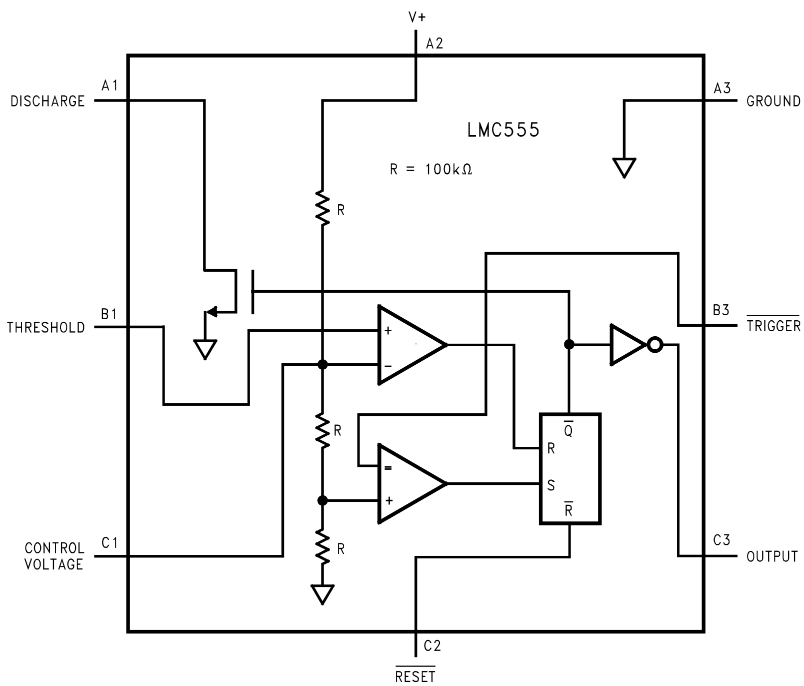


Figure 2. 8-Bump DSBGA Top View (Bump Side Down)

Table 1. Pin Descriptions

Pin Name	Package Pin Numbers	
	8-Pin SOIC, VSSOP, and PDIP	8-Bump DSBGA
GND	1	A3
$\overline{\text{Trigger}}$	2	B3
Output	3	C3
$\overline{\text{Reset}}$	4	C2
Control Voltage	5	C1
Threshold	6	B1
Discharge	7	A1
V^+	8	A2

These devices have limited built-in ESD protection. The leads should be shorted together or the device placed in conductive foam during storage or handling to prevent electrostatic damage to the MOS gates.

Absolute Maximum Ratings (1)(2)(3)

Supply Voltage, V^+	15V
Input Voltages, V_{TRIG}, V_{RES}, V_{CTRL}, V_{THRESH}	−0.3V to V_S + 0.3V
Output Voltages, V_O, V_{DIS}	15V
Output Current I_O, I_{DIS}	100 mA
Storage Temperature Range	−65°C to +150°C
Soldering specification for PDIP package:	
Soldering (10 seconds)	260°C
Soldering specification for all other packages: see product folder at www.ti.com and http://www.ti.com/lit/SNOA549	

(1) Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is functional, but do not ensure specific performance limits. Electrical Characteristics state DC and AC electrical specifications under particular test conditions which ensure specific performance limits. This assumes that the device is within the Operating Ratings. Specifications are not ensured for parameters where no limit is given, however, the typical value is a good indication of device performance.
(2) See AN-1112 (SNVA009) for DSBGA considerations.
(3) If Military/Aerospace specified devices are required, please contact the TI Sales Office/Distributors for availability and specifications.

Operating Ratings (1)(2)

Temperature Range	
LMC555IM	−40°C to +125°C
LMC555CM/MM/N/TP	−40°C to +85°C
Thermal Resistance (θ_{JA}) (1)	
SOIC, 8-Pin	169°C/W
VSSOP, 8-Pin	225°C/W
PDIP, 8-Pin	111°C/W
8-Bump DSBGA	220°C/W
Maximum Allowable Power Dissipation @25°C	
PDIP-8	1126 mW
SOIC-8	740 mW
VSSOP-8	555 mW
8-Bump DSBGA	568 mW

(1) Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is functional, but do not ensure specific performance limits. Electrical Characteristics state DC and AC electrical specifications under particular test conditions which ensure specific performance limits. This assumes that the device is within the Operating Ratings. Specifications are not ensured for parameters where no limit is given, however, the typical value is a good indication of device performance.
(2) See AN-1112 (SNVA009) for DSBGA considerations.

Electrical Characteristics[(1)] [(2)]

Test Circuit, T = 25°C, all switches open, $\overline{\text{RESET}}$ to V_S unless otherwise noted

Parameter		Test Conditions	Min	Typ	Max	Units (Limits)
I_S	Supply Current	V_S = 1.5V V_S = 5V V_S = 12V		50 100 150	150 250 400	µA
V_{CTRL}	Control Voltage	V_S = 1.5V V_S = 5V V_S = 12V	0.8 2.9 7.4	1.0 3.3 8.0	1.2 3.8 8.6	V
V_{DIS}	Discharge Saturation Voltage	V_S = 1.5V, I_{DIS} = 1 mA V_S = 5V, I_{DIS} = 10 mA		75 150	150 300	mV
V_{OL}	Output Voltage (Low)	V_S = 1.5V, I_O = 1 mA V_S = 5V, I_O = 8 mA V_S = 12V, I_O = 50 mA		0.2 0.3 1.0	0.4 0.6 2.0	V
V_{OH}	Output Voltage (High)	V_S = 1.5V, I_O = −0.25 mA V_S = 5V, I_O = −2 mA V_S = 12V, I_O = −10 mA	1.0 4.4 10.5	1.25 4.7 11.3		V
V_{TRIG}	Trigger Voltage	V_S = 1.5V V_S = 12V	0.4 3.7	0.5 4.0	0.6 4.3	V
I_{TRIG}	Trigger Current	V_S = 5V		10		pA
V_{RES}	Reset Voltage	V_S = 1.5V [(3)] V_S = 12V	0.4 0.4	0.7 0.75	1.0 1.1	V
I_{RES}	Reset Current	V_S = 5V		10		pA
I_{THRESH}	Threshold Current	V_S = 5V		10		pA
I_{DIS}	Discharge Leakage	V_S = 12V		1.0	100	nA
t	Timing Accuracy	SW 2, 4 Closed V_S = 1.5V V_S = 5V V_S = 12V	 0.9 1.0 1.0	 1.1 1.1 1.1	 1.25 1.20 1.25	ms
$\Delta t/\Delta V_S$	Timing Shift with Supply	V_S = 5V ± 1V		0.3		%/V
$\Delta t/\Delta T$	Timing Shift with Temperature	V_S = 5V		75		ppm/°C
f_A	Astable Frequency	SW 1, 3 Closed, V_S = 12V	4.0	4.8	5.6	kHz
f_{MAX}	Maximum Frequency	Max. Freq. Test Circuit, V_S = 5V		3.0		MHz
t_R, t_F	Output Rise and Fall Times	Max. Freq. Test Circuit V_S = 5V, C_L = 10 pF		15		ns
t_{PD}	Trigger Propagation Delay	V_S = 5V, Measure Delay from Trigger to Output		100		ns

(1) All voltages are measured with respect to the ground pin, unless otherwise specified.
(2) Absolute Maximum Ratings indicate limits beyond which damage to the device may occur. Operating Ratings indicate conditions for which the device is functional, but do not ensure specific performance limits. Electrical Characteristics state DC and AC electrical specifications under particular test conditions which ensure specific performance limits. This assumes that the device is within the Operating Ratings. Specifications are not ensured for parameters where no limit is given, however, the typical value is a good indication of device performance.
(3) If the $\overline{\text{RESET}}$ pin is to be used at temperatures of −20°C and below V_S is required to be 2.0V or greater.

Timer

NE/SA/SE555/SE555C

DESCRIPTION

The 555 monolithic timing circuit is a highly stable controller capable of producing accurate time delays, or oscillation. In the time delay mode of operation, the time is precisely controlled by one external resistor and capacitor. For a stable operation as an oscillator, the free running frequency and the duty cycle are both accurately controlled with two external resistors and one capacitor. The circuit may be triggered and reset on falling waveforms, and the output structure can source or sink up to 200mA.

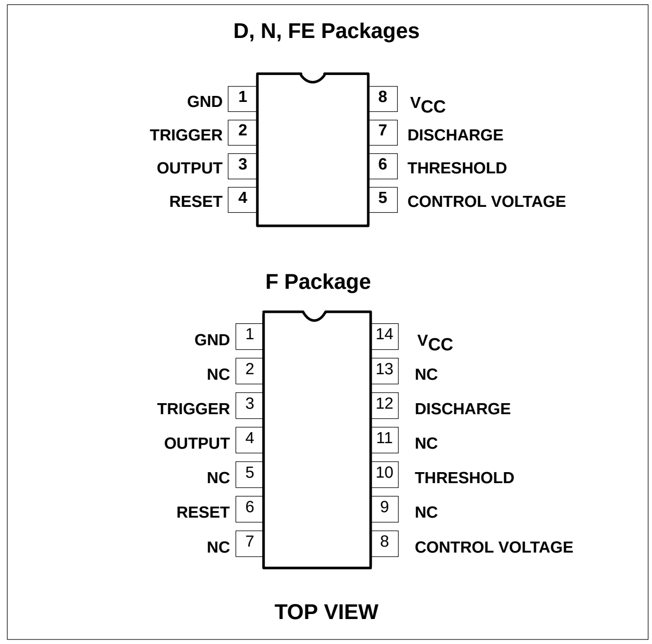
FEATURES

- Turn-off time less than 2µs
- Max. operating frequency greater than 500kHz
- Timing from microseconds to hours
- Operates in both astable and monostable modes
- High output current
- Adjustable duty cycle
- TTL compatible
- Temperature stability of 0.005% per °C

APPLICATIONS

- Precision timing
- Pulse generation
- Sequential timing
- Time delay generation
- Pulse width modulation

PIN CONFIGURATIONS

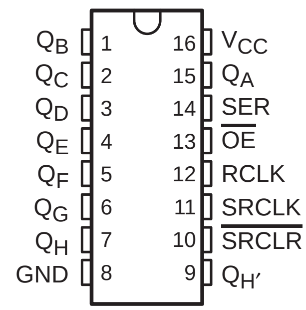


ORDERING INFORMATION

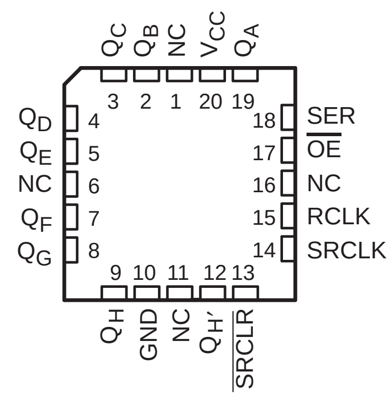
DESCRIPTION	TEMPERATURE RANGE	ORDER CODE	DWG #
8-Pin Plastic Small Outline (SO) Package	0 to +70°C	NE555D	0174C
8-Pin Plastic Dual In-Line Package (DIP)	0 to +70°C	NE555N	0404B
8-Pin Plastic Dual In-Line Package (DIP)	-40°C to +85°C	SA555N	0404B
8-Pin Plastic Small Outline (SO) Package	-40°C to +85°C	SA555D	0174C
8-Pin Hermetic Ceramic Dual In-Line Package (CERDIP)	-55°C to +125°C	SE555CFE	
8-Pin Plastic Dual In-Line Package (DIP)	-55°C to +125°C	SE555CN	0404B
14-Pin Plastic Dual In-Line Package (DIP)	-55°C to +125°C	SE555N	0405B
8-Pin Hermetic Cerdip	-55°C to +125°C	SE555FE	
14-Pin Ceramic Dual In-Line Package (CERDIP)	0 to +70°C	NE555F	0581B
14-Pin Ceramic Dual In-Line Package (CERDIP)	-55°C to +125°C	SE555F	0581B
14-Pin Ceramic Dual In-Line Package (CERDIP)	-55°C to +125°C	SE555CF	0581B

- **8-Bit Serial-In, Parallel-Out Shift**
- **High-Current 3-State Outputs Can Drive up to 15 LSTTL Loads**
- **Shift Register Has Direct Clear**
- **Package Options Include Plastic Small-Outline (D) and Ceramic Flat (W) Packages, Ceramic Chip Carriers (FK), and Standard Plastic (N) and Ceramic (J) 300-mil DIPs**

SN54HC595 . . . J OR W PACKAGE
SN74HC595 . . . D OR N PACKAGE
(TOP VIEW)



SN54HC595 . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

description

The ’HC595 contain an 8-bit serial-in, parallel-out shift register that feeds an 8-bit D-type storage register. The storage register has parallel 3-state outputs. Separate clocks are provided for both the shift and storage register. The shift register has a direct overriding clear ($\overline{\text{SRCLR}}$) input, serial (SER) input, and serial outputs for cascading.

Both the shift register clock (RCLK) and storage register clock (SRCLK) are positive-edge triggered. If both clocks are connected together, the shift register is always one clock pulse ahead of the storage register.

The SN54HC595 is characterized for operation over the full military temperature range of –55°C to 125°C. The SN74HC595 is characterized for operation from –40°C to 85°C.

MOTOROLA

SN54/74LS48

BCD TO 7-SEGMENT DECODER

The SN54/74LS48 is a BCD to 7-Segment Decoder consisting of NAND gates, input buffers and seven AND-OR-INVERT gates. Seven NAND gates and one driver are connected in pairs to make BCD data and its complement available to the seven decoding AND-OR-INVERT gates. The remaining NAND gate and three input buffers provide lamp test, blanking input/ripple-blanking input for the LS48.

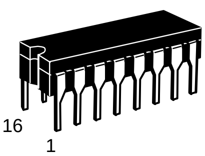
The circuit accepts 4-bit binary-coded-decimal (BCD) and, depending on the state of the auxiliary inputs, decodes this data to drive other components. The relative positive logic output levels, as well as conditions required at the auxiliary inputs, are shown in the truth tables.

The LS48 circuit incorporates automatic leading and/or trailing edge zero-blanking control (RBI and RBO). Lamp Test (LT) may be activated any time when the BI/RBO node is HIGH. Both devices contain an overriding blanking input (BI) which can be used to control the lamp intensity by varying the frequency and duty cycle of the BI input signal or to inhibit the outputs.

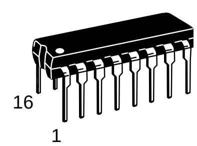
- Lamp Intensity Modulation Capability (BI/RBO)
- Internal Pull-Ups Eliminate Need for External Resistors
- Input Clamp Diodes Eliminate High-Speed Termination Effects

BCD TO 7-SEGMENT DECODER

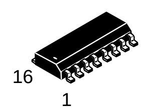
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

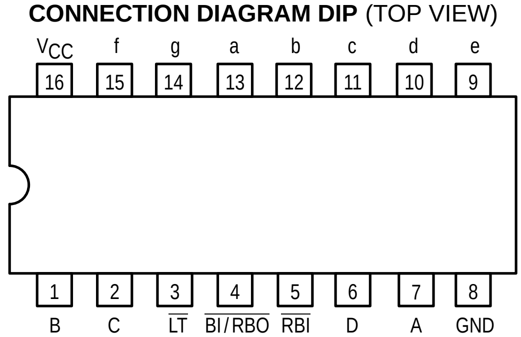


D SUFFIX
SOIC
CASE 751B-03

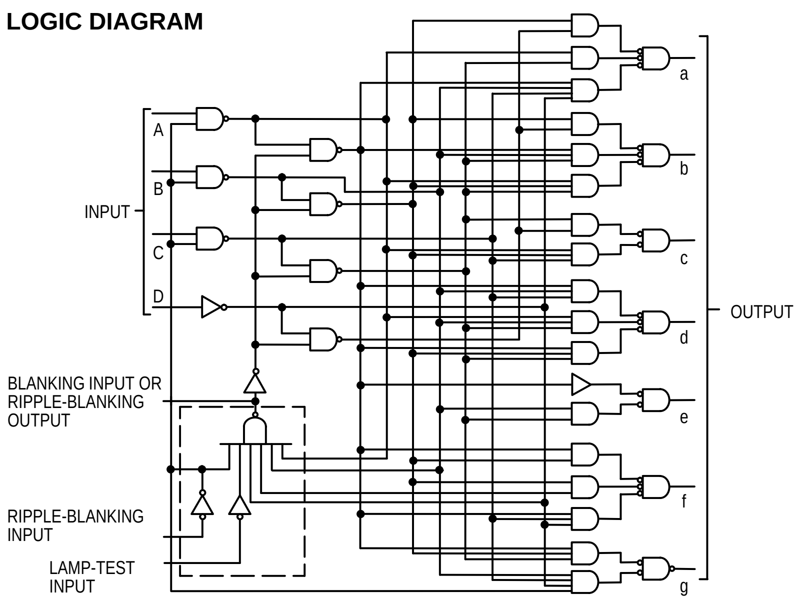
ORDERING INFORMATION

SN54LSXXJ	Ceramic
SN74LSXXN	Plastic
SN74LSXXD	SOIC

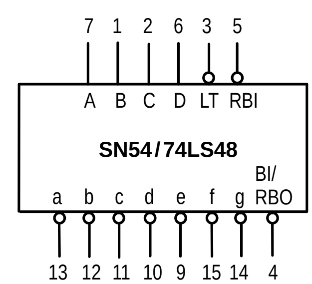
CONNECTION DIAGRAM DIP (TOP VIEW)



LOGIC DIAGRAM



LOGIC SYMBOL

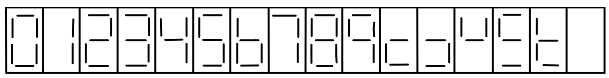


V_{CC} = PIN 16
GND = PIN 8

PIN NAMES | **LOADING** (Note a)

		HIGH	LOW
A, B, C, D	BCD Inputs	0.5 U.L.	0.25 U.L.
$\overline{RBI}$	Ripple-Blanking (Active Low) Input	0.5 U.L.	0.25 U.L.
$\overline{LT}$	Lamp-Test (Active Low) Input	0.5 U.L.	0.25 U.L.
$\overline{BI/RBO}$	Blanking Input or Ripple-	0.5 U.L.	0.75 U.L.
	Blanking Output (Active Low)	1.2 U.L.	2(1) U.L.
$\overline{BI}$	Blanking (Active Low) Input	0.5 U.L.	0.25 U.L.
		Open-Collector	3.75 (1.25) U.L. (48)

NOTES:
a) Unit Load (U.L.) = 40 µA HIGH/1.6 mA LOW
b) Outut current measured at V_{OUT} = 0.5 V
Output LOW drive factor is SN54LS/74LS48: 1.25 U.L. for Military (54), 3.75 U.L. for Commercial (74).



0 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15

NUMERICAL DESIGNATIONS — RESULTANT DISPLAYS

TRUTH TABLE
SN54/74LS48

DECIMAL OR FUNCTION	$\overline{LT}$	$\overline{RBI}$	D	C	B	A	$\overline{BI/RBO}$	a	b	c	d	e	f	g	NOTE
	INPUTS							OUTPUTS							
0	H	H	L	L	L	L	H	H	H	H	H	H	H	L	1
1	H	X	L	L	L	H	H	L	H	H	L	L	L	L	1
2	H	X	L	L	H	L	H	H	H	L	H	H	L	H	
3	H	X	L	L	H	H	H	H	H	H	H	L	L	H	
4	H	X	L	H	L	L	H	L	H	H	L	L	H	H	
5	H	X	L	H	L	H	H	H	L	H	H	L	H	H	
6	H	X	L	H	H	L	H	L	L	H	H	H	H	H	
7	H	X	L	H	H	H	H	H	H	H	L	L	L	L	
8	H	X	H	L	L	L	H	H	H	H	H	H	H	H	
9	H	X	H	L	L	H	H	H	H	H	L	L	H	H	
10	H	X	H	L	H	L	H	L	L	L	H	H	L	H	
11	H	X	H	L	H	H	H	L	L	H	H	L	L	H	
12	H	X	H	H	L	L	H	L	H	L	L	L	H	H	
13	H	X	H	H	L	H	H	H	L	L	H	L	H	H	
14	H	X	H	H	H	L	H	L	L	L	H	H	H	H	
15	H	X	H	H	H	H	H	L	L	L	L	L	L	L	
$\overline{BI}$	X	X	X	X	X	X	L	L	L	L	L	L	L	L	2
$\overline{RBI}$	H	L	L	L	L	L	L	L	L	L	L	L	L	L	3
$\overline{LT}$	L	X	X	X	X	X	H	H	H	H	H	H	H	H	4

NOTES:
(1) $\overline{BI/RBO}$ is wired-AND logic serving as blanking input ($\overline{BI}$) and/or ripple-blanking output ($\overline{RBO}$). The blanking out ($\overline{BI}$) must be open or held at a HIGH level when output functions 0 through 15 are desired, and ripple-blanking input ($\overline{RBI}$) must be open or at a HIGH level if blanking of a decimal 0 is not desired. X=input may be HIGH or LOW.
(2) When a LOW level is applied to the blanking input (forced condition) all segment outputs go to a LOW level, regardless of the state of any other input condition.
(3) When ripple-blanking input ($\overline{RBI}$) and inputs A, B, C, and D are at LOW level, with the lamp test input at HIGH level, all segment outputs go to a HIGH level and the ripple-blanking output ($\overline{RBO}$) goes to a LOW level (response condition).
(4) When the blanking input/ripple-blanking output ($\overline{BI/RBO}$) is open or held at a HIGH level, and a LOW level is applied to lamp-test input, all segment outputs go to a LOW level.

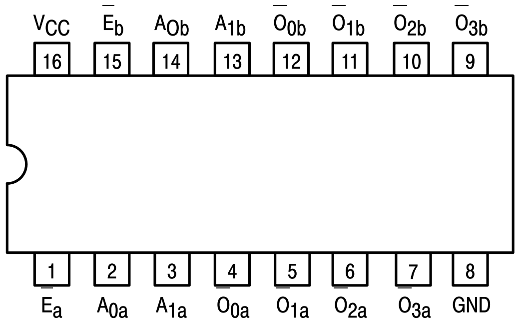
MOTOROLA

DUAL 1-OF-4 DECODER/ DEMULTIPLEXER

The LSTTL/MSI SN54/74LS139 is a high speed Dual 1-of-4 Decoder/Demultiplexer. The device has two independent decoders, each accepting two inputs and providing four mutually exclusive active LOW Outputs. Each decoder has an active LOW Enable input which can be used as a data input for a 4-output demultiplexer. Each half of the LS139 can be used as a function generator providing all four minterms of two variables. The LS139 is fabricated with the Schottky barrier diode process for high speed and is completely compatible with all Motorola TTL families.

- Schottky Process for High Speed
- Multifunction Capability
- Two Completely Independent 1-of-4 Decoders
- Active Low Mutually Exclusive Outputs
- Input Clamp Diodes Limit High Speed Termination Effects
- ESD > 3500 Volts

CONNECTION DIAGRAM DIP (TOP VIEW)



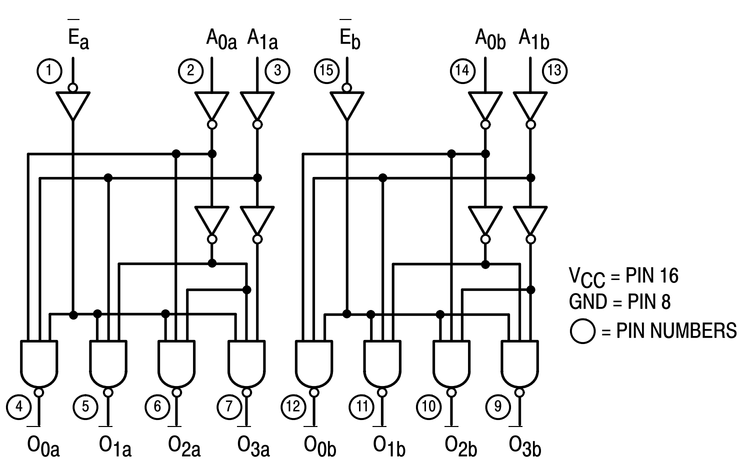
NOTE:
The Flatpak version
has the same pinouts
(Connection Diagram) as
the Dual In-Line Package.

PIN NAMES / **LOADING** (Note a)

		HIGH	LOW
A_0, A_1	Address Inputs	0.5 U.L.	0.25 U.L.
$\overline{E}$	Enable (Active LOW) Input	0.5 U.L.	0.25 U.L.
$\overline{O}_0$–$\overline{O}_3$	Active LOW Outputs (Note b)	10 U.L.	5 (2.5) U.L.

NOTES:
a) 1 TTL Unit Load (U.L.) = 40 µA HIGH/1.6 mA LOW.
b) The Output LOW drive factor is 2.5 U.L. for Military (54) and 5 U.L. for Commercial (74) Temperature Ranges.

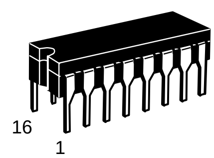
LOGIC DIAGRAM



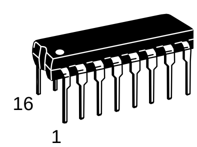
SN54/74LS139

DUAL 1-OF-4 DECODER/ DEMULTIPLEXER

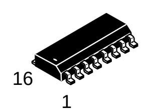
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

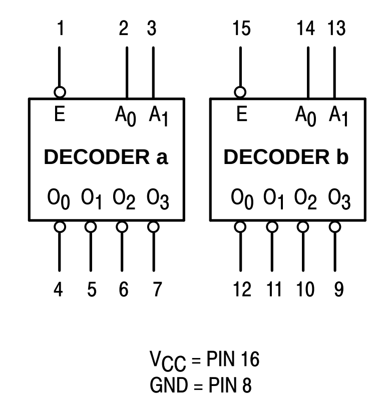


D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

SN54LSXXXJ	Ceramic
SN74LSXXXN	Plastic
SN74LSXXXD	SOIC

LOGIC SYMBOL



FUNCTIONAL DESCRIPTION

The LS139 is a high speed dual 1-of-4 decoder/demultiplexer fabricated with the Schottky barrier diode process. The device has two independent decoders, each of which accept two binary weighted inputs (A_0, A_1) and provide four mutually exclusive active LOW outputs ($\overline{O}_0$ – $\overline{O}_3$). Each decoder has an active LOW Enable ($\overline{E}$). When $\overline{E}$ is HIGH all outputs are forced HIGH. The enable can be used as the data input for a 4-output demultiplexer application.

Each half of the LS139 generates all four minterms of two variables. These four minterms are useful in some applications, replacing multiple gate functions as shown in Fig. a, and thereby reducing the number of packages required in a logic network.

TRUTH TABLE

INPUTS			OUTPUTS			
$\overline{E}$	A_0	A_1	$\overline{O}_0$	$\overline{O}_1$	$\overline{O}_2$	$\overline{O}_3$
H	X	X	H	H	H	H
L	L	L	L	H	H	H
L	H	L	H	L	H	H
L	L	H	H	H	L	H
L	H	H	H	H	H	L

H = HIGH Voltage Level
L = LOW Voltage Level
X = Don't Care

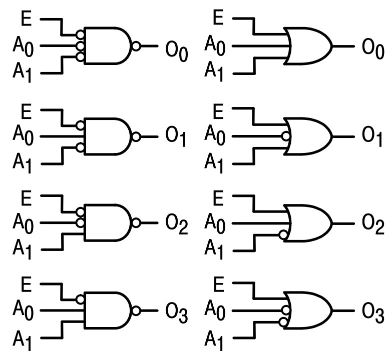


Figure a

GUARANTEED OPERATING RANGES

Symbol	Parameter		Min	Typ	Max	Unit
V_{CC}	Supply Voltage	54 74	4.5 4.75	5.0 5.0	5.5 5.25	V
T_A	Operating Ambient Temperature Range	54 74	–55 0	25 25	125 70	°C
I_{OH}	Output Current — High	54, 74			–0.4	mA
I_{OL}	Output Current — Low	54 74			4.0 8.0	mA

SDLS058

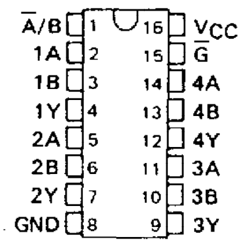
SN54157, SN54LS157, SN54LS158, SN54S157, SN54S158, SN74157, SN74LS157, SN74LS158, SN74S157, SN74S158 QUADRUPLE 2-LINE TO 1-LINE DATA SELECTORS/MULTIPLEXERS

MARCH 1974 – REVISED MARCH 1988

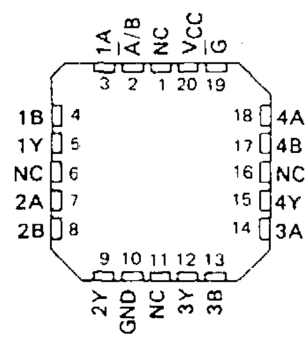
- Buffered Inputs and Outputs
- Three Speed/Power Ranges Available

TYPES	TYPICAL AVERAGE PROPAGATION TIME	TYPICAL POWER DISSIPATION
'157	9 ns	150 mW
'LS157	9 ns	49 mW
'S157	5 ns	250 mW
'LS158	7 ns	24 mW
'S158	4 ns	195 mW

SN54157, SN54LS157, SN54S157, SN54LS158, SN54S158 . . . J OR W PACKAGE
SN74157 . . . N PACKAGE
SN74LS157, SN74S157, SN74LS158, SN74S158 . . . D OR N PACKAGE
(TOP VIEW)



SN54LS157, SN54S157, SN54LS158, SN54S158 . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

applications

- Expand Any Data Input Point
- Multiplex Dual Data Buses
- Generate Four Functions of Two Variables (One Variable Is Common)
- Source Programmable Counters

description

These monolithic data selectors/multiplexers contain inverters and drivers to supply full on-chip data selection to the four output gates. A separate strobe input is provided. A 4-bit word is selected from one of two sources and is routed to the four outputs. The '157, 'LS157, and 'S157 present true data whereas the 'LS158 and 'S158 present inverted data to minimize propagation delay time.

FUNCTION TABLE

INPUTS				OUTPUT Y	
STROBE $\overline{G}$	SELECT $\overline{A}/B$	A	B	'157, 'LS157, 'S157	'LS158 'S158
H	X	X	X	L	H
L	L	L	X	L	H
L	L	H	X	H	L
L	H	X	L	L	H
L	H	X	H	H	L

H = high level, L = low level, X = irrelevant

absolute maximum ratings over operating free-air temperature range (unless otherwise noted)

Supply voltage, V_{CC} (See Note 1)	7 V
Input voltage: '157, 'S158	5.5 V
'LS157, 'LS158	7 V
Operating free-air temperature range: SN54'	−55°C to 125°C
SN74'	0°C to 70°C
Storage temperature range	−65°C to 150°C

NOTE 1: Voltage values are with respect to network ground terminal.

SN54/74LS173A

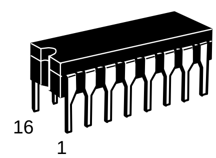
4-BIT D-TYPE REGISTER WITH 3-STATE OUTPUTS

The SN54/74LS173A is a high-speed 4-Bit Register featuring 3-state outputs for use in bus-organized systems. The clock is fully edge-triggered allowing either a load from the D inputs or a hold (retain register contents) depending on the state of the Input Enable Lines ($\overline{IE}_1$, $\overline{IE}_2$). A HIGH on either Output Enable line (OE_1, OE_2) brings the output to a high impedance state without affecting the actual register contents. A HIGH on the Master Reset (MR) input resets the Register regardless of the state of the Clock (CP), the Output Enable (OE_1, OE_2) or the Input Enable ($\overline{IE}_1$, $\overline{IE}_2$) lines.

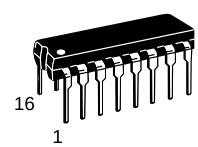
- Fully Edge-Triggered
- 3-State Outputs
- Gated Input and Output Enables
- Input Clamp Diodes Limit High-Speed Termination Effects

4-BIT D-TYPE REGISTER WITH 3-STATE OUTPUTS

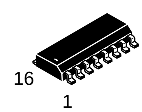
LOW POWER SCHOTTKY



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08

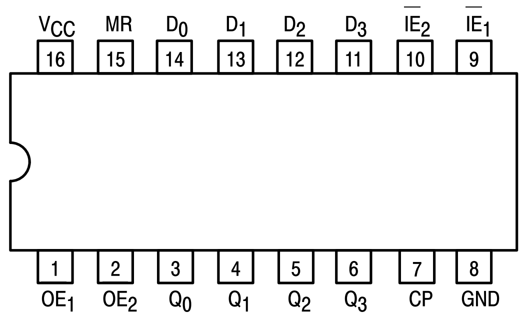


D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

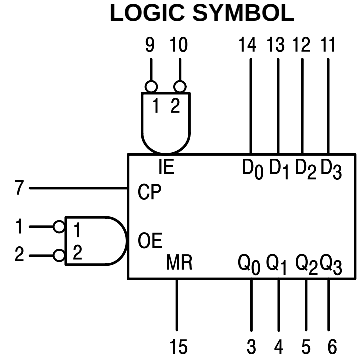
SN54LSXXXJ	Ceramic
SN74LSXXXN	Plastic
SN74LSXXXD	SOIC

CONNECTION DIAGRAM DIP (TOP VIEW)



NOTE:
The Flatpak version has the same pinouts (Connection Diagram) as the Dual In-Line Package.

LOGIC SYMBOL

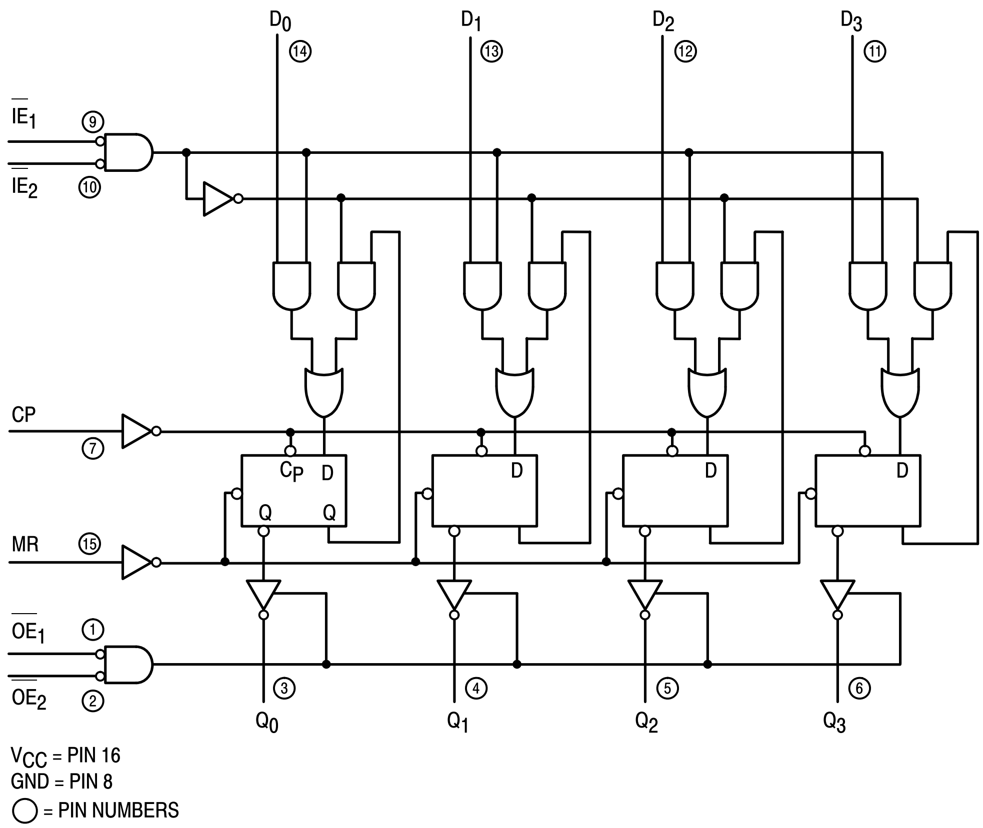


V_{CC} = PIN 16
GND = PIN 8

PIN NAMES		**LOADING** (Note a) HIGH	LOW
$D_0 – D_3$	Data Inputs	0.5 U.L.	0.25 U.L.
$\overline{IE}_1 – \overline{IE}_2$	Input Enable (Active LOW)	0.5 U.L.	0.25 U.L.
$OE_1 – OE_2$	Output Enable (Active LOW) Inputs	0.5 U.L.	0.25 U.L.
CP	Clock Pulse (Active HIGH Going Edge) Input	0.5 U.L.	0.25 U.L.
MR	Master Reset Input (Active HIGH)	0.5 U.L.	0.25 U.L.
$Q_0 – Q_3$	Outputs (Note b)	65 (25) U.L.	15 (7.5) U.L.

NOTES:
a. 1 TTL Unit Load (U.L.) = 40 µA HIGH/1.6 mA LOW.
b. The Output LOW drive factor is 2.5 U.L. for Military (54) and 5 U.L. for Commercial (74) Temperature Ranges.

LOGIC DIAGRAM



TRUTH TABLE

MR	CP	$\overline{IE}_1$	$\overline{IE}_2$	D_n	Q_n
H	x	x	x	x	L
L	L	x	x	x	Q_n
L	↑	H	x	x	Q_n
L	↑	x	H	x	Q_n
L	↑	L	L	L	L
L	↑	L	L	H	H

When either OE_1, or OE_2 are HIGH, the output is in the off state (High Impedance); however this does not affect the contents or sequential operation of the register.

H = HIGH Voltage Level
L = LOW Voltage Level
X = Immaterial

GUARANTEED OPERATING RANGES

Symbol	Parameter		Min	Typ	Max	Unit
V_{CC}	Supply Voltage	54 74	4.5 4.75	5.0 5.0	5.5 5.25	V
T_A	Operating Ambient Temperature Range	54 74	–55 0	25 25	125 70	°C
I_{OH}	Output Current — High	54 74			–1.0 –2.6	mA
I_{OL}	Output Current — Low	54 74			12 24	mA

MOTOROLA

4-BIT BINARY FULL ADDER WITH FAST CARRY

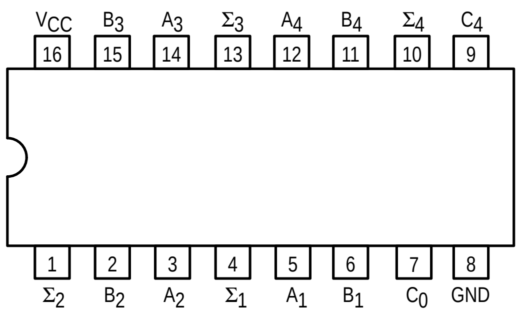
SN54/74LS283

4-BIT BINARY FULL ADDER WITH FAST CARRY

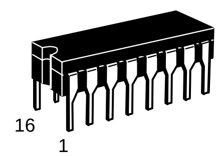
LOW POWER SCHOTTKY

The SN54/74LS283 is a high-speed 4-Bit Binary Full Adder with internal carry lookahead. It accepts two 4-bit binary words (A_1–A_4, B_1–B_4) and a Carry Input (C_0). It generates the binary Sum outputs (Σ_1–Σ_4) and the Carry Output (C_4) from the most significant bit. The LS283 operates with either active HIGH or active LOW operands (positive or negative logic).

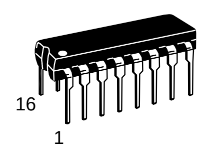
CONNECTION DIAGRAM DIP (TOP VIEW)



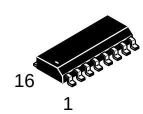
NOTE:
The Flatpak version has the same pinouts (Connection Diagram) as the Dual In-Line Package.



J SUFFIX
CERAMIC
CASE 620-09



N SUFFIX
PLASTIC
CASE 648-08



D SUFFIX
SOIC
CASE 751B-03

ORDERING INFORMATION

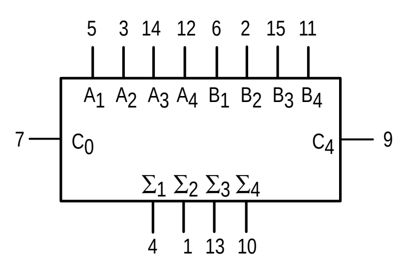
SN54LSXXXJ	Ceramic
SN74LSXXXN	Plastic
SN74LSXXXD	SOIC

PIN NAMES | **LOADING** (Note a)

		HIGH	LOW
A_1–A_4	Operand A Inputs	1.0 U.L.	0.5 U.L.
B_1–B_4	Operand B Inputs	1.0 U.L.	0.5 U.L.
C_0	Carry Input	0.5 U.L.	0.25 U.L.
Σ_1–Σ_4	Sum Outputs (Note b)	10 U.L.	5 (2.5) U.L.
C_4	Carry Output (Note b)	10 U.L.	5 (2.5) U.L.

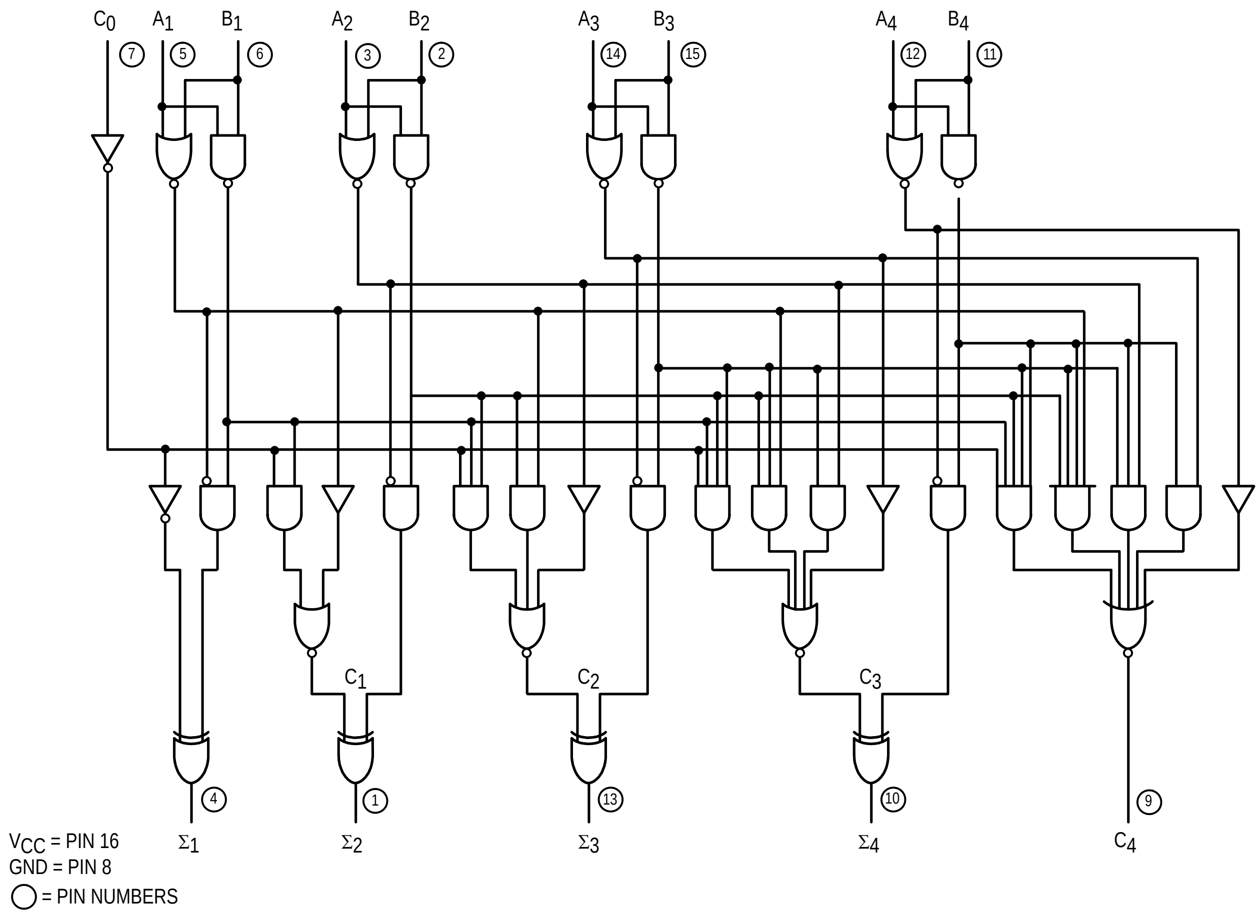
NOTES:
a) 1 TTL Unit Load (U.L.) = 40 µA HIGH/1.6 mA LOW.
b) The Output LOW drive factor is 2.5 U.L. for Military (54) and 5 U.L. for Commercial (74) Temperature Ranges.

LOGIC SYMBOL



V_{CC} = PIN 16
GND = PIN 8

LOGIC DIAGRAM



FUNCTIONAL DESCRIPTION

The LS283 adds two 4-bit binary words (A plus B) plus the incoming carry. The binary sum appears on the sum outputs ($\Sigma_1 - \Sigma_4$) and outgoing carry (C4) outputs.

$$C_0 + (A_1 + B_1) + 2(A_2 + B_2) + 4(A_3 + B_3) + 8(A_4 + B_4) = \Sigma_1 + 2\Sigma_2 + 4\,\Sigma_3 + 8\,\Sigma_4 + 16C_4$$

Where: (+) = plus

Due to the symmetry of the binary add function the LS283 can be used with either all inputs and outputs active HIGH (positive logic) or with all inputs and outputs active LOW (negative logic). Note that with active HIGH inputs, Carry Input can not be left open, but must be held LOW when no carry in is intended.

Example:

	C_0	A_1	A_2	A_3	A_4	B_1	B_2	B_3	B_4	Σ_1	Σ_2	Σ_3	Σ_4	C_4	
logic levels	L	L	H	L	H	H	L	L	H	H	H	L	L	H	
Active HIGH	0	0	1	0	1	1	0	0	1	1	1	0	0	1	(10+9=19)
Active LOW	1	1	0	1	0	0	1	1	0	0	0	1	1	0	(carry+5+6=12)

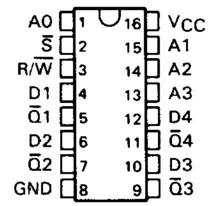
Interchanging inputs of equal weight does not affect the operation, thus C_0, A_1, B_1, can be arbitrarily assigned to pins 7, 5 or 3.

64-BIT RANDOM-ACCESS MEMORIES

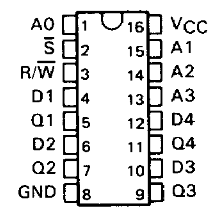
D2417, SEPTEMBER 1980 – REVISED FEBRUARY 1985

- **Organized as 16 Words of Four Bits Each**
- **Choice of Buffered 3-State or Open-Collector outputs**
- **Choice of Noninverted or Inverted Outputs**
- **Typical Access Time . . . 50 ns**

SN54LS189A, SN54LS289A . . . J PACKAGE
SN74LS189A, SN74LS289A . . . J OR N PACKAGE
(TOP VIEW)



SN54LS219A, SN54LS319A . . . J PACKAGE
SN74LS219A, SN74LS319A . . . J OR N PACKAGE
(TOP VIEW)



description

These monolithic TTL memories feature Schottky clamping for high performance and a fast chip-select access time to enhance decoding at the system level. A three-state output version and an open-collector-output version are offered for both of the logic choices. A three-state output offers the convenience of an open-collector output with the speed of a totem-pole output; it can be bus-connected to other similar outputs, yet it retains the fast rise time characteristic of the TTL totem-pole output. An open-collector output offers the capability of direct interface with a data line having a passive pull-up.

write cycle

Information to be stored in the memory is written into the selected address location when the chip-select ($\overline{S}$) and the write-enable (R/$\overline{W}$) inputs are low. While the write-enable input is low, the memory outputs are off (three-state = Hi-Z, open-collector = high). When a number of outputs are bus-connected, this off state neither loads nor drives the data bus; however, it permits the bus line to be driven by other active outputs or a passive pull-up.

read cycle

Information stored in the memory (see function table for input/output phase relationship) is available at the outputs when the write-enable input is high and the chip-select input is low. When the chip-select input is high, the outputs will be off.

FUNCTION TABLE

FUNCTION	INPUTS		OUTPUTS			
	CHIP SELECT	WRITE ENABLE	'LS189A	'LS289A	'LS219A	'LS319A
Write	L	L	Z	Off	Z	Off
Read	L	H	Complement of Data Entered	Complement of Data Entered	Data Entered	Data Entered
Inhibit	H	X	Z	Off	Z	Off

H = high level, L = low level, X = irrelevant, Z = high impedance

5

RAMs

TEXAS INSTRUMENTS
POST OFFICE BOX 225012 • DALLAS, TEXAS 75265

- **Package Options Include Plastic "Small Outline" Packages, Ceramic Chip Carriers, and Plastic and Ceramic DIPs**
- **Dependable Texas Instruments Quality and Reliability**

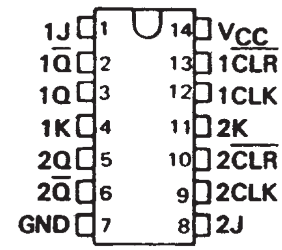
description

The '107 contain two independent J-K flip-flops with individual J-K, clock, and direct clear inputs. The '107 is a positive pulse-triggered flip-flop. The J-K input data is loaded into the master while the clock is high and transferred to the slave and the outputs on the high-to-low clock transistion. For these devices the J and K inputs must be stable while the clock is high.

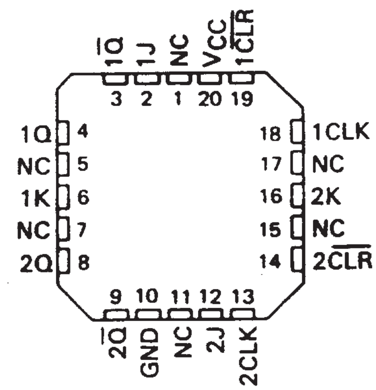
The 'LS107A contain two independent negative-edge-triggered flip-flops. The J and K inputs must be stable prior to the high-to-low clock transition for predictable operation. When the clear is low, it overrides the clock and data inputs forcing the Q output low and the $\overline{Q}$ output high.

The SN54107 and the SN54LS107A are characterized for operation over the full military temperature range of −55°C to 125°C. The SN74107 and the SN74LS107A are characterized for operation from 0°C to 70°C.

SN54107, SN54LS107A . . . J PACKAGE
SN74107 . . . N PACKAGE
SN74LS107A . . . D OR N PACKAGE
(TOP VIEW)



SN54LS107A . . . FK PACKAGE
(TOP VIEW)



NC – No internal connection

'107
FUNCTION TABLE

INPUTS				OUTPUTS	
$\overline{CLR}$	CLK	J	K	Q	$\overline{Q}$
L	X	X	X	L	H
H	⎍	L	L	Q_0	$\overline{Q}_0$
H	⎍	H	L	H	L
H	⎍	L	H	L	H
H	⎍	H	H	TOGGLE	

'LS107A
FUNCTION TABLE

INPUTS				OUTPUTS	
$\overline{CLR}$	CLK	J	K	Q	$\overline{Q}$
L	X	X	X	L	H
H	↓	L	L	Q_0	$\overline{Q}_0$
H	↓	H	L	H	L
H	↓	L	H	L	H
H	↓	H	H	TOGGLE	
H	H	X	X	Q_0	$\overline{Q}_0$

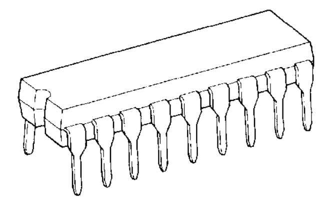
TOSHIBA Bipolar Digital Integrated Circuit Silicon Monolithic

ULN2803APG,ULN2803AFWG,ULN2804APG,ULN2804AFWG (Manufactured by Toshiba Malaysia)

8ch Darlington Sink Driver

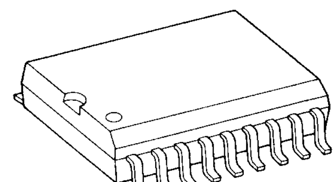
The ULN2803APG / AFWG Series are high−voltage, high−current darlington drivers comprised of eight NPN darlington pairs.
All units feature integral clamp diodes for switching inductive loads.
Applications include relay, hammer, lamp and display (LED) drivers.
The suffix (G) appended to the part number represents a Lead (Pb)-Free product.

ULN2803APG
ULN2804APG



DIP18-P-300-2.54F

ULN2803AFWG
ULN2804AFWG



SOL18-P-300-1.27

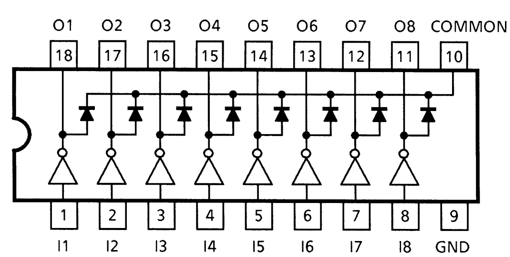
Weight
DIP18−P−300−2.54F: 1.478 g (Typ.)
SOL18−P−300−1.27 : 0.48 g (Typ.)

Features

- Output current (single output)
  500 mA (Max.)
- High sustaining voltage output
  50 V (Min.)
- Output clamp diodes
- Inputs compatible with various types of logic.
- Package Type−APG : DIP−18pin
- Package Type−AFWG : SOL−18pin

Type	Input Base Resistor	Designation
ULN2803APG / AFWG	2.7 kΩ	TTL, 5 V CMOS
ULN2804APG / AFWG	10.5 kΩ	6~15 V PMOS, CMOS

Pin Connection (top view)



Recommended Operating Conditions (Ta = −40~85°C)

Characteristic		Symbol	Test Condition	Min	Typ.	Max	Unit
Output sustaining voltage		$V_{CE\ (SUS)}$		0	—	50	V
Output current	APG	I_{OUT}	T_{pw} = 25 ms, Duty = 10%, 8 Circuits	0	—	347	mA / ch
			T_{pw} = 25 ms, Duty = 50%, 8 Circuits	0	—	123	
	AFWG		T_{pw} = 25 ms, Duty = 10%, 8 Circuits	0	—	268	
			T_{pw} = 25 ms, Duty = 50%, 8 Circuits	0	—	90	
Input voltage		V_{IN}		0	—	30	V
Input voltage (Output on)	ULN2803A	$V_{IN\ (ON)}$		3.5	—	30	V
	ULN2804A			8	—	30	
Clamp diode reverse voltage		V_R		—	—	50	V
Clamp diode forward current		I_F		—	—	400	mA
Power dissipation	APG	P_D	Ta = 85°C	—	—	0.76	W
	AFWG		Ta = 85°C (Note)	—	—	0.48	

Note: On Glass Epoxy PCB (75 × 114 × 1.6 mm Cu 20%)